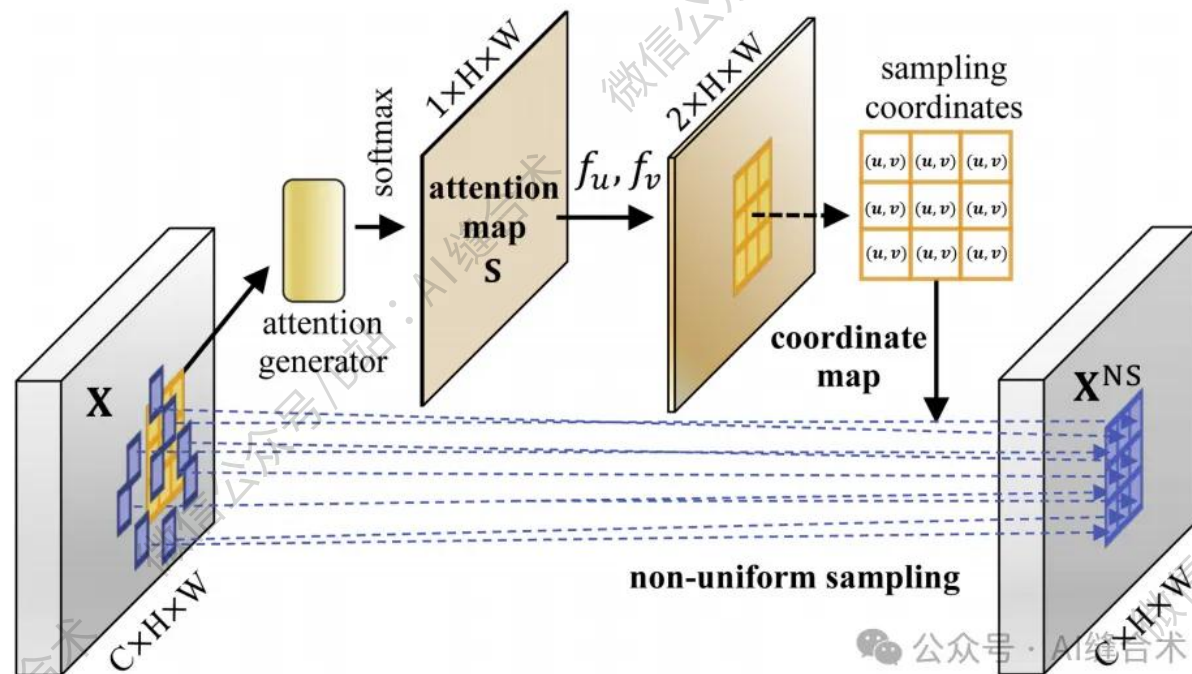


核心速览



论文题目: Spatial Frequency Modulation for Semantic Segmentation

中文题目: 基于空间频率调制的语义分割方法

论文链接:

<https://ieeexplore.ieee.org/abstract/document/11095993>

所属单位: 北京理工大学等

核心速览:

本文提出空间频率调制（SFM）框架，通过调制-解调机制解决语义分割中高频信息在降采样过程中的混叠问题。该框架包含自适应重采样（ARS）和多尺度自适应上采样（MSAU）两个轻量级模块，前者将高频特征调制为低频以避免混叠，后者在恢复时解调并重建高频细节，实现端到端训练且兼容 CNN 与 Transformer 架构。

https://mp.weixin.qq.com/s/YCY2oMM8AkwBzdLRI_JGHQ

```
ARS(
  (attention_generator): AttentionGenerator(
    (daconv): DAConv()
    (psp): PSPModule(
      (stages): ModuleList(
        (0): Sequential(
          (0): AdaptiveAvgPool2d(output_size=1)
          (1): Conv2d(64, 16, kernel_size=(1, 1), stride=(1, 1))
          (2): GELU(approximate='none')
        )
        (1): Sequential(
          (0): AdaptiveAvgPool2d(output_size=2)
          (1): Conv2d(64, 16, kernel_size=(1, 1), stride=(1, 1))
          (2): GELU(approximate='none')
        )
        (2): Sequential(
          (0): AdaptiveAvgPool2d(output_size=3)
          (1): Conv2d(64, 16, kernel_size=(1, 1), stride=(1, 1))
          (2): GELU(approximate='none')
        )
        (3): Sequential(
          (0): AdaptiveAvgPool2d(output_size=7)
          (1): Conv2d(64, 16, kernel_size=(1, 1), stride=(1, 1))
          (2): GELU(approximate='none')
        )
      )
    )
    (fuse): Sequential(
      (0): Conv2d(128, 16, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
      (1): GELU(approximate='none')
      (2): Conv2d(16, 1, kernel_size=(1, 1), stride=(1, 1))
    )
    (coordinate_mapper): CoordinateMapper()
  )
  input_tensor_shape : torch.Size([2, 64, 128, 128])
  output_tensor_shape : torch.Size([2, 64, 128, 128])
)
```

哔哩哔哩/微信公众号：AI缝合术，独家整理！